

The idea of the ecological footprint is helpful for understanding how everyday lifestyle choices affect the planet. It measures how much land and water it takes to produce things we use, from the food to shopping. It also looks at how much land is needed to offset all the waste we create, especially carbon emissions. This helps quantify how our personal habits affect global climate change by showing how much space is needed for extracting resources and absorbing any waste. By tracking all of this, the Global Footprint Network calculator provides a holistic picture of an individual's direct impact on climate change.

The assessment showed that we would need 6.5 Earths to survive if everyone lived like me. My ecological footprint was 10.6 global hectares, with my carbon footprint making up the majority of that. The figures page showed that my two largest areas of resource consumption are travel and food. What surprised me the most was how destructive my eating habits are on the environment, despite how "healthy" they are.

The travel category was the biggest reason why my footprint was so high. My daily drives are efficient since my apartment is very close to my job site, meaning I drive less than 70 miles a week. But, traveling back and forth to visit my parents in Honolulu twice a year adds up fast. Also considering flights to engineering conferences, that amount of fuel heavily increased my footprint.

Food was my second highest impact area. My diet primarily consists of ground beef, rice, and cheese. The BBC Diet Calculator provided a better breakdown of how this impacts emissions. The beef I eat contributes about 5,641 kilograms to my annual greenhouse gas emissions, which is the equivalent of driving 14,392 miles. Cheese adds another 704 kilograms of emissions with another 1,798 miles. Since I only drive about 4,000 miles a year, my diet has the equivalent impact of driving 4x that. This calculator showed me that beef and dairy

have a hidden ecological cost due to the land, water, and methane involved, which is not something I had ever considered before.

On top of that, the assessment showed me the impact of my shopping habits. I regularly shop online to buy items like saxophone reeds and engineering software. Buying things constantly means I am always dependent on a global supply chain that requires raw materials, packaging, and shipping.

When looking at where I need to improve, the “perfect solution” would be to stop flying entirely and become a vegan who only eats local food. Since this is unrealistic, I plan to focus on simple changes for now. Not flying to see my parents is not a reasonable change for me at 19 in this current stage of my life. Also, cutting out meat and cheese overnight is not a healthy decision. For travel, I can commit to keeping my daily driving low even after my internship ends since I am an RA and can walk to class instead of driving. For my diet, I can swap out beef for different proteins two or three days a week. Finally, I can find alternative engineering programs that the university provides and utilize reusable plastic reeds.

The biggest obstacles to making these changes are habitual ones, as well as time and convenience. The fact that the academic and professional world heavily relies on air travel is a challenge that is hard to fix on my own. These tools did a great job showing me how much my daily habits connect to global ecology. Even small, easy changes in my day-to-day life are necessary to reduce my burden on the planet.